

Daniel DeKerlegand

AI Researcher & Senior Engineer

New Orleans, LA

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SUMMARY

Senior engineer and PhD candidate in computer science with a concentration in AI/ML. 10+ years of professional software experience in AI/ML, API, web, mobile, and distributed systems engineering. Doctoral research assistant conducting research in AI/ML and NLP with a focus on hybrid and neuro-symbolic AI architectures, procedural generation, and social physics.

EXPERIENCE

Evolve Media AI

Jun 2021 – Present

Director of Artificial Intelligence · New Orleans, LA

- Architected production microservices platform with 8 services (Python Flask, Node.js Express, React) deployed on Kubernetes (AWS EKS, GKE, AKS) using Infrastructure-as-Code (Terraform), implementing GraphQL API, WebSocket subscriptions, and Redis pub/sub for real-time collaborative applications serving thousands of users.
- Developed sophisticated AI orchestration system using Google Gemini 2.5 Pro with 14+ function calling tools for multi-modal content analysis (video, audio, text), implementing machine learning preference systems with weighted scoring algorithms for personalized recommendations and scalable video intelligence pipelines with parallel processing architectures.
- Built comprehensive full-stack applications including real-time video editor with React/Remotion, cross-platform mobile app (React Native/Expo) with biometric authentication, and REST/GraphQL APIs with JWT authentication, PostgreSQL database, automated deployment pipelines, and monitoring achieving 99.9% uptime.
- Designed intuitive AI-powered interfaces with conversational assistants, multi-turn dialogue support, real-time progress feedback, and personalized user experiences combining symbolic reasoning with neural generation for adaptive content creation responding to user behavior patterns.

University of New Orleans

Aug 2019 – Present

Artificial Intelligence Researcher · New Orleans, LA

- Conducted multi-phase research program spanning 5 implemented systems demonstrating hybrid AI effectiveness across education, entertainment, and worldbuilding domains, integrating neuro-symbolic architectures combining first-order predicate logic with transformer-based language models (GPT-4), validated through three user studies (n=69 total participants) with dissertation research cited 358+ times in scholarly literature.
- Developed comprehensive neuro-symbolic AI system (Insimul) with bidirectional information flow between symbolic social state and statistical language generation, implementing rule extraction and compilation pipelines, knowledge graphs with neural embeddings (FAISS), and advanced prompt engineering using DSPy achieving 95%+ consistency in multi-generational social simulations with 40-200+ procedurally-generated characters.
- Built full-stack TypeScript application with React frontend, Node.js/Express backend, MongoDB persistence, and Prolog rule engine supporting real-time collaborative editing via WebSocket, implementing social physics engines for emergent narrative in Unity/Unreal Engine 5 with MetaHumans integration and VR language learning systems achieving comparable performance to human tutors with significantly higher confidence gains.
- Designed rigorous evaluation frameworks including controlled experiments with statistical modeling (mixed-effects models, R/lme4), multi-generational demographic simulation algorithms, collaborative knowledge engineering workshops with domain experts, and procedural content generation systems for culturally-grounded virtual characters with automatic scaling and production deployment handling 69+ participants.

Torsh

Dec 2015 – Jun 2021

Lead Engineer · New Orleans, LA

- Oversaw, from conception to production, the complete rewrite of legacy iOS and Android apps as a single React Native app. Directed other engineers and operationalized organizational engineering standards.
- Architected and implemented app navigation, authentication, data persistence, and core UI components. Built bespoke React Native libraries in Objective C and Java for audio/video domain-specific tasks like Bluetooth connectivity and camera management.
- Designed, developed, and deployed GraphQL API in Node.js. Integrated Apollo client into application, adding an offline-first local caching system.
- Implemented critical fullstack features in a Meteor/Node.js web application: a pub-sub task queuing system, audio-video processing services, an internal reporting library of data analytics, a PDF annotation tool, an OpenWRT video chat tool, and a CMS.

EDUCATION

University of New Orleans New Orleans, LA

Doctor of Philosophy, Computer ScienceExpected May 2026

Master of Science, Computer Science2018

Bachelor of Arts, English Language & Literature · Bachelor of Arts, Film/Cinema/Video Studies2010

University of Arizona Tucson, AZ

Master of Fine Arts, Creative Writing2013

CORE SKILLS

Symbolic AI & Logic

- SWI-Prolog
- Ensemble
- Insimul DSL
- Datalog
- ASP
- CLIPS
- Pyke
- Knowledge Graphs
- OWL
- SPARQL
- Social Physics

AI/ML Tooling

- PyTorch
- Keras
- TensorFlow
- MLFlow
- Jupyter
- HuggingFace

LLM Inference, Training & Optimization

- Open R1
- MinGPT
- Megatron LM
- Finetune Transformer LM
- DSPy
- Retrieval-Augmented Generation
- FAISS
- ChromaDB
- Pinecone
- Vector Embeddings
- Semantic Search
- Prompt Engineering
- Few-Shot Learning
- Chain-of-Thought Prompting
- LoRA/ControlNet

Agentic AI

- Model Context Protocol
- Agent-to-Agent
- Agent Development Kit
- Google Vertex Agents
- LangChain
- LangGraph
- CrewAI
- AutoGen
- BabyAGI
- AgentGPT
- Semantic Kernel
- ReAct
- Dify
- Flowise

Speech & Audio

- ffmpeg
- Whisper STT
- Coqui TTS
- Piper TTS
- ElevenLabs
- Google Cloud TTS
- espeak-ng
- Oculus Lip Sync
- SALSA
- Azure Speech Services
- Amazon Polly
- Wav2Vec
- DeepSpeech
- Tacotron
- VITS
- Phoneme Analysis

Game Engines & VR

- Unity
- Unity Sentis
- Unreal Engine 5
- MetaHumans
- Oculus SDK
- OpenXR
- SteamVR
- XR Interaction Toolkit
- Godot
- WebXR
- A-Frame
- Spatial Audio SDK

Statistical Model Architectures

- RNN
- CNN
- LSTM
- GAN
- BERT
- Transformer
- T5
- Attention Mechanisms
- Encoder-Decoder Models
- VAE
- ResNet
- U-Net
- Graph Neural Networks
- Diffusion Models

Procedural Generation

- Tracery
- Perlin Noise
- Wave Function Collapse
- L-Systems
- Markov Chains
- Context-Free Grammars
- Cellular Automata
- Noise Functions (Simplex, Worley)
- PCG Algorithms
- Rule-Based Generation

Data Analysis & Research

- R
- lme4
- Montreal Forced Aligner
- NVivo
- Python (NumPy, Pandas, SciPy)
- Matplotlib
- ggplot2
- SPSS
- Jupyter Notebooks
- Tableau
- Statistical Modeling
- Mixed-Effects Models

Languages & Frameworks

- Python
- Rust
- Go
- TypeScript
- Node.js
- C#/.NET
- Clojure
- Java
- React
- Vue
- React Native
- Swift
- Kotlin

DevOps/MLOps, Data & Cloud-Native

- Terraform
- Helm
- Chef
- Ansible
- Kubernetes
- Jenkins
- Jenkins X
- CircleCI
- AWS
- GCP
- Grafana
- Sentry
- Datadog
- Kibana
- Prometheus
- GitHub Actions
- MongoDB
- PostgreSQL
- Drizzle ORM
- MySQL
- SQLite
- Redis
- Neo4j
- Prisma
- TypeORM
- Supabase